

Name: Key

/ 25

Please circle your section: 901 (Gossell) 902 (Gossell) 903 (Graves-McCleary)

No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. [12 points] Solve the following limits. Show your work and use correct limit notation.

$$\text{a. } \lim_{x \rightarrow 2} \frac{x^3 - 2x^2}{6 - 3x} = \lim_{x \rightarrow 2} \frac{x^2(x-2)}{-3(x-2)} = \lim_{x \rightarrow 2} \frac{x^2}{-3} = \frac{2^2}{-3} = \boxed{-\frac{4}{3}}$$

$$\text{b. } \lim_{x \rightarrow 1} \frac{\frac{x}{2} - \frac{2}{x}}{2x - 4} = \frac{\frac{1}{2} - \frac{2}{1}}{2(1) - 4} = \frac{-\frac{3}{2}}{-2} = \boxed{\frac{3}{4}}$$

$$\begin{aligned} \text{c. } \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x+4} - 2)(\sqrt{x+4} + 2)}{x(\sqrt{x+4} + 2)} = \lim_{x \rightarrow 0} \frac{x+4-4}{x(\sqrt{x+4} + 2)} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+4} + 2)} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+4} + 2} = \frac{1}{2+2} = \boxed{\frac{1}{4}} \end{aligned}$$

2. [8 points] Consider the following piecewise function:

$$g(x) = \begin{cases} \frac{x^2 - 4}{x + 2}, & x < -2 \\ \sqrt{4 - x^2}, & -2 \leq x \leq 2 \\ x^2 - 4, & x > 2 \end{cases}$$

a. Is $g(x)$ continuous at $x = -2$? Justify your answer using limits.

No.

$$\lim_{x \rightarrow -2^-} g(x) = \lim_{x \rightarrow -2^-} \frac{x^2 - 4}{x + 2} = \lim_{x \rightarrow -2^-} \frac{(x - 2)(x + 2)}{x + 2} = \lim_{x \rightarrow -2^-} (x - 2) = -2 - 2 = -4$$

$$\lim_{x \rightarrow -2^+} g(x) = \lim_{x \rightarrow -2^+} \sqrt{4 - x^2} = \sqrt{4 - (-2)^2} = \sqrt{0} = 0$$

So $\lim_{x \rightarrow -2} g(x)$ DNE.

b. Is $g(x)$ continuous at $x = 2$? Justify your answer using limits.

Yes

$$\lim_{x \rightarrow 2^-} g(x) = \lim_{x \rightarrow 2^-} \sqrt{4 - x^2} = \sqrt{4 - 2^2} = \sqrt{0} = 0$$

$$\lim_{x \rightarrow 2^+} g(x) = \lim_{x \rightarrow 2^+} (x^2 - 4) = 2^2 - 4 = 0$$

So $\lim_{x \rightarrow 2} g(x) = 0 = g(2)$.

$$g(2) = 0$$

3. [5 points] Use the Intermediate Value Theorem to show that the graph of $f(x) = 3^x - x - 20$ passes through the x -axis between $x = 2$ and $x = 3$.

$$f(2) = 3^2 - 2 - 20 = -13 < 0$$

$$f(3) = 3^3 - 3 - 20 = 4 > 0$$

Since $f(x)$ is continuous and since $f(2) < 0 < f(3)$, by the IVT there must be an x -value in $(2, 3)$ with $f(x) = 0$.